International Journal of Advanced Artificial Intelligence Research (IJAAIR)
eISSN: 3087-4270
Volume03 Issue09 | Page No: - 17-23

Received: 30 Jun 2026, Revised: 15 Jul 2026, Accepted: 25 Aug 2026, Published: 09 Sep 2026

DOI: https://doi.org/10.55640/ijaair-v03i09-03

# Dynamic Risk-Based Access Control for Autonomous Agentic AI Systems: Architecture, Policy Enforcement, and Security Evaluation

**Nethika Perera [1], Kavindu Jayasinghe[2]**
[1] Artificial Intelligence Research Specialist, Sri Lanka
[2] AI Systems Research Analyst, Sri Lanka

**ABSTRACT:** Autonomous agentic artificial intelligence (AI) systems increasingly operate as decision-making entities capable of selecting tools, invoking services, processing information, and executing actions with limited human intervention. This autonomy creates an access-control problem that cannot be adequately addressed through static permission models because the security significance of an action depends on contextual variables such as requested resource, operational state, transaction sensitivity, delegation history, environmental conditions, and estimated risk. This paper proposes a dynamic risk-based access-control architecture for autonomous agentic AI systems that combines identity verification, contextual risk assessment, policy evaluation, least-privilege enforcement, adaptive authorization, and continuous security monitoring. The proposed methodology conceptualizes authorization as a dynamic decision function rather than a one-time permission assignment. The architecture introduces a multi-stage policy-enforcement pipeline in which an agent's identity and delegated authority are evaluated together with action context and risk indicators before access is granted, constrained, escalated, or denied. The research further develops a security-evaluation framework covering authorization accuracy, privilege containment, policy consistency, decision latency, adaptability, and resistance to privilege escalation. The risk-adaptive authorization concept is particularly relevant because recent work demonstrates that contextual authorization can combine workload identity, policy evaluation, risk scoring, constrained tokens, and differentiated approval levels for autonomous agents (Bhushan, Pappu, Jadav, and Jaiswal, 2026). Drawing on optimization-oriented and data-driven system research, the proposed framework emphasizes continuous recalibration rather than fixed authorization thresholds. Analytical findings indicate that dynamic authorization can improve security resilience and reduce unnecessary privilege exposure while introducing computational and governance trade-offs. The study contributes an integrated conceptual architecture and evaluation model for implementing risk-sensitive access control in autonomous AI environments.

**KEYWORDS:** Agentic AI, Dynamic Access Control, Risk-Based Authorization, Autonomous Agents, Policy Enforcement, Adaptive Security, Least Privilege, AI Security, Authorization Architecture, Security Evaluation.

## 1. Introduction

The transition from conventional AI applications toward autonomous agentic AI systems changes the security assumptions underlying access control. A conventional software application generally executes within a comparatively stable permission boundary, whereas an autonomous agent can interpret objectives, select tools, communicate with external services, delegate subtasks, and initiate multiple actions during a single operational episode. Consequently, authorization must address not only whether an entity is recognized but also whether a particular action should be permitted under the circumstances in which it is requested.

Static access-control mechanisms are insufficient for this environment because identical permissions can have substantially different security consequences depending on context. For example, an agent authorized to retrieve customer information may legitimately perform a low-risk read operation during routine processing but create significant exposure if it attempts bulk extraction during an abnormal session. The permission itself has not changed; the risk associated with its use has changed. Dynamic authorization therefore requires a decision mechanism capable of evaluating identity, action, resource, context, and risk simultaneously.

This problem can be positioned within a broader theoretical perspective of data-driven decision optimization. Danish demonstrates how machine-learning techniques can be incorporated into the

modeling and optimization of complex energy systems, illustrating the importance of transforming heterogeneous operational information into actionable decisions (Danish, 2024). Similarly, dynamic authorization requires transformation of heterogeneous security signals into an operational access decision. The distinction is that, in security, optimization cannot simply maximize operational efficiency; it must preserve mandatory security constraints.

Recent research directly focused on agentic systems strengthens this argument. Bhushan et al. propose risk-adaptive authorization that combines workload identity, token exchange, policy evaluation, AI-based risk scoring, sender-constrained tokens, and differentiated approval mechanisms for autonomous agents (Bhushan, Pappu, Jadav, and Jaiswal, 2026). Their work demonstrates the feasibility of treating risk as an authorization input rather than as a post-event monitoring variable.

The objective of this research is therefore to develop a structured architecture for dynamic risk-based access control in autonomous agentic AI systems. The study focuses on five objectives: defining the architecture of contextual authorization; formalizing risk-sensitive policy decisions; establishing mechanisms for privilege containment; developing an evaluation methodology; and analyzing the trade-off between security, adaptability, and authorization latency.

The significance of the proposed framework lies in its attempt to connect autonomous decision-making with deterministic security enforcement. The framework does not assume that an AI-generated risk assessment should independently grant authority. Instead, risk assessment functions as an input to a policy-constrained authorization mechanism. This distinction is essential because an uncertain or manipulated AI signal should not automatically become an authorization capability.

## 2. Literature Review

### 2.1 Data-Driven Decision Optimization

Danish presents a framework for modeling and optimizing data-driven energy systems through machine learning, providing a useful theoretical analogy for dynamic security decisions (Danish, 2024). The central transferable principle is that complex operational environments require models capable of incorporating changing conditions rather than relying exclusively on predetermined decisions. In an agentic AI environment, contextual security information can similarly be treated as a dynamic decision space.

However, authorization differs fundamentally from conventional optimization. Energy-system optimization may seek efficiency, stability, or cost minimization, whereas access control includes non-negotiable security constraints. Therefore, a risk score should not be interpreted as an unrestricted optimization objective. Instead, it should influence policy selection within a bounded security model.

The solar-energy literature further demonstrates the importance of systematic evaluation under variable operating conditions. Hayat et al. analyze power-generation challenges and future requirements within solar-energy systems, highlighting how technological performance must be assessed in relation to changing operational constraints (Hayat et al., 2019). The same principle applies to autonomous AI authorization: security performance must be evaluated under different risk states rather than under a single nominal condition.

### 2.2 Optimization Under Complex System Conditions

Pfahl's studies of heliostat development and heliostat concepts demonstrate the importance of optimizing system configurations while accounting for practical constraints (Pfahl, 2017; Pfahl, 2014). In access-control architecture, the equivalent problem is determining how permissions should be configured so that legitimate autonomy is preserved without unnecessarily expanding the attack surface.

Rizvi et al. provide a classification of heliostat-field layouts and optimization techniques, illustrating how alternative configurations can be compared systematically rather than evaluated as isolated designs (Rizvi et al., 2021). This principle informs the proposed access-control framework by treating authorization policies as configurable structures that can be evaluated according to security and operational performance.

Wei et al. similarly investigate heliostat-field layout design as a structured optimization problem (Wei et al., 2010). The transferable methodological insight is that system-level performance depends not only on individual components but also on their arrangement and interaction. For agentic AI, authorization quality similarly depends on the interaction among identity, context, policy, risk, and enforcement components.

### 2.3 Risk, Variability, and Adaptive Control

He et al. analyze non-uniform solar-flux characteristics, challenges, and possible solutions in concentrated solar power systems (He et al., 2019). Although the application domain is different, the work reinforces the importance of designing systems for non-uniform and changing conditions. A security architecture designed around a fixed threat model can exhibit similar weaknesses when operating conditions change.

Hamanah et al. provide a comprehensive survey of solar-power tower drives and demonstrate the

complexity of interconnected control components (Hamanah et al., 2023). This supports a systems-level interpretation of access control in which authorization cannot be treated as an isolated identity check. Autonomous agents operate across interconnected resources, and a local authorization decision can affect downstream services.

Varun et al. evaluate alternative configurations through SOLTRACE-based analysis, illustrating the usefulness of simulation and comparative testing for complex technical systems (Varun et al., 2020). A comparable simulation-oriented approach is appropriate for evaluating dynamic authorization policies before deployment.

Coventry and Pye examine heliostat cost-reduction alternatives and emphasize the need to reconsider system design as technological conditions evolve (Coventry and Pye, 2014). In agentic security, adaptive policies similarly need periodic recalibration as agents, resources, threats, and operational patterns change.

### 2.4 Research Gap

The principal gap identified from the supplied literature is not an absence of optimization methodologies but the absence of direct integration between those methodologies and autonomous-agent authorization. The supplied studies demonstrate modeling, classification, optimization, comparative evaluation, and adaptation in complex systems, but they do not establish an access-control architecture for autonomous AI agents.

The agentic-AI authorization study supplied by the user directly addresses this missing security dimension by introducing risk-adaptive authorization and differentiated approval levels (Bhushan, Pappu, Jadav, and Jaiswal, 2026). The present research extends that conceptual direction by emphasizing dynamic policy recalibration, contextual risk aggregation, formal policy constraints, and a broader security-evaluation framework.

## 3. Methodology

### 3.1 Research Design

This study adopts a design-oriented architectural methodology. The proposed system is developed as a layered authorization framework rather than as a single access-control algorithm. The design follows six stages:

1. agent identity establishment;
2. contextual information collection;
3. risk estimation;
4. policy evaluation;
5. adaptive enforcement; and
6. continuous security evaluation.

The methodology treats authorization as a constrained decision function:

A=F(I,C,R,P,D)A = F(I, C, R, P, D)

where II represents agent identity and trust state, CC represents contextual attributes, RR represents estimated risk, PP represents applicable policy, and DD represents delegation constraints. The output AA can be permit, permit-with-restrictions, escalate, or deny.

The structure follows the principle that complex systems should be evaluated through interacting variables rather than isolated components. This methodological orientation is consistent with the optimization perspective evident in the supplied energy-system literature (Rizvi et al., 2021; Wei et al., 2010).

### 3.2 Agent Identity and Delegation Layer

The first layer establishes the identity of the autonomous agent. Identity must include more than a static identifier. The authorization engine should consider agent instance, parent agent, delegated scope, execution environment, session state, and credential freshness.

Delegation is represented as:

Schild⊆SparentS_{child} \subseteq S_{parent}

where SS denotes the authorized scope set. This monotonic delegation constraint prevents a subordinate agent from obtaining privileges beyond those delegated by its parent.

For example, an orchestration agent may be authorized to create a temporary analytical sub-agent with permission to read anonymized data. The sub-agent should not automatically inherit permissions to modify databases or export sensitive records. This limits privilege propagation across autonomous agent hierarchies.

### 3.3 Context Collection

The second layer gathers contextual attributes associated with each authorization request. Relevant variables include requested resource, requested operation, data sensitivity, transaction value, location or execution environment, time, session behavior, historical activity, delegation depth, and anomaly indicators.

A contextual vector may be represented as:

C={c1,c2,…,cn}C = \{c_1,c_2,\ldots,c_n\}

The purpose of contextualization is to distinguish nominally identical operations that possess different

security implications.

For example, a read request for ten public records may be low risk, whereas the same agent requesting ten million sensitive records within a short interval should trigger a substantially different authorization response.

### 3.4 Dynamic Risk Model

The risk engine converts contextual indicators into a normalized risk score:

R=∑i=1nwixiR = \sum_{i=1}^{n}w_i x_i

where xix_i denotes a normalized risk factor and wiw_i represents its relative importance.

A more advanced implementation can incorporate nonlinear interactions:

R=f(X,H,T,D)R = f(X, H, T, D)

where XX represents current context, HH historical behavior, TT threat indicators, and DD delegation characteristics.

The critical design principle is that the risk model does not directly issue permission. Instead, the risk output is passed to a deterministic policy engine. This separation reduces the possibility that an erroneous AI prediction becomes an unrestricted security decision.

Bhushan et al. similarly position AI-derived risk as an input to policy-controlled authorization rather than as an independent authorization mechanism, demonstrating the practical value of combining adaptive risk assessment with deterministic enforcement (Bhushan, Pappu, Jadav, and Jaiswal, 2026).

### 3.5 Policy Evaluation

The policy engine evaluates the request against mandatory security constraints. A policy can be expressed as:

Permit=IdentityValid∧ScopeValid∧ResourceAllowed∧Risk<RmaxPermit = IdentityValid \land ScopeValid \land ResourceAllowed \land Risk<R_{max}

Additional conditions can define adaptive actions:

Decision={Permit,R<θ1Restricted,θ1≤R<θ2Escalate,θ2≤R<θ3Deny,R≥θ3Decision = \begin{cases} Permit, & R < \theta_1\\ Restricted, & \theta_1 \leq R < \theta_2\\ Escalate, & \theta_2 \leq R < \theta_3\\ Deny, & R \geq \theta_3 \end{cases}

This four-level model is preferable to a binary allow/deny mechanism for autonomous systems because it allows low-risk actions to proceed automatically while reserving increasingly strong controls for elevated-risk activity.

### 3.6 Adaptive Enforcement

The enforcement layer applies the policy decision to the requested action.

A low-risk operation may receive normal access. A moderate-risk operation may receive a restricted scope or reduced transaction limit. A high-risk operation may require approval from another supervisory agent or human operator. A critical-risk operation should be denied.

This architecture recognizes that excessive restriction can reduce the practical value of autonomous AI. Therefore, security should be proportional to risk rather than uniformly restrictive.

The conceptual approach is consistent with the broader idea of selecting system configurations according to operational requirements rather than applying one universal configuration (Pfahl, 2014; Coventry and Pye, 2014).

### 3.7 Continuous Evaluation

Authorization must not end when access is initially granted. Continuous monitoring should determine whether subsequent behavior remains consistent with the authorization context.

For example, an agent may initially be approved to retrieve 100 records. If it subsequently attempts to retrieve 100,000 records, the authorization state should be reevaluated. This transforms access control from a one-time decision into a continuous control loop.

The resulting model is:

Accesst→Monitort→Reassesst+1→Enforcet+1Access_t \rightarrow Monitor_t \rightarrow Reassess_{t+1} \rightarrow Enforce_{t+1}

Such feedback-oriented reasoning is important in variable operating environments, where static assumptions can become obsolete.

### 3.8 Security-Evaluation Framework

The proposed architecture is evaluated across six dimensions:

Authorization accuracy: percentage of legitimate requests correctly permitted and malicious requests correctly rejected.

Privilege containment: ability to prevent agents from exceeding delegated permissions.

Policy consistency: degree to which equivalent contexts receive consistent authorization outcomes.

Decision latency: time required to complete identity, risk, policy, and enforcement evaluation.

Adaptability: ability to change decisions as contextual risk changes.

Security resilience: resistance to privilege escalation, abnormal behavior, policy manipulation, and compromised-agent activity.

Simulation should include low-, medium-, high-, and critical-risk scenarios. Multiple configurations should be compared to determine whether adaptive policies achieve better security-efficiency balance than static authorization.

## 4. Results

The analytical evaluation of the proposed architecture indicates that dynamic risk-based authorization provides a stronger security model for autonomous agents than a static permission structure. The primary finding is that authorization quality improves when the decision function considers contextual risk in addition to identity and predefined permissions. Under a static model, an authenticated agent can continue using an authorized capability even when its operational behavior changes. Under the proposed architecture, the authorization state can be recalculated as contextual variables change.

The first major finding concerns privilege containment. The delegation constraint Schild⊆SparentS_{child}\subseteq S_{parent} provides a simple but powerful invariant for controlling autonomous-agent hierarchies. If a parent agent delegates a narrowly defined task, the child cannot legitimately expand that authority through ordinary authorization requests. This reduces the potential blast radius of compromised or malfunctioning agents.

The second finding concerns risk proportionality. A multi-level decision model avoids the limitations of binary authorization. Low-risk actions can receive rapid automated decisions, while increasingly sensitive actions can invoke restrictions, supervisory approval, or denial. This is particularly important for autonomous systems because requiring human approval for every action would substantially reduce operational efficiency.

The third finding concerns policy stability. Risk scores alone are unsuitable as authorization decisions because probabilistic assessments may be uncertain. Combining the risk signal with deterministic policy constraints produces a layered model in which AI contributes adaptability without becoming the final security authority. This principle aligns closely with the risk-adaptive authorization approach proposed by Bhushan et al. (2026).

The fourth finding concerns continuous security. Reassessment allows an agent's authorization state to respond to behavioral changes. This is more resilient than an authorization model in which permissions remain unchanged throughout a session.

The fifth finding concerns evaluation methodology. The supplied optimization literature demonstrates the value of comparing alternative configurations rather than evaluating one design in isolation. Pfahl's work on heliostat concepts and Rizvi et al.'s classification of optimization approaches illustrate how configuration-level comparison can expose trade-offs that component-level analysis misses (Pfahl, 2014; Rizvi et al., 2021). Applied to authorization, this implies that security evaluation should compare threshold settings, policy structures, risk models, and enforcement strategies.

The principal operational trade-off is latency. More contextual attributes, risk computation, and policy evaluation can increase authorization overhead. Therefore, the optimal design is not the one with the highest number of security checks but the one that provides sufficient protection while maintaining acceptable response time. This mirrors the broader engineering principle that complex systems must balance competing performance objectives (Varun et al., 2020).

Overall, the findings support a dynamic authorization model in which risk acts as a policy input, delegation boundaries constrain privilege, and continuous monitoring enables adaptive enforcement.

## 5. Discussion

The findings indicate that dynamic authorization represents a conceptual shift from permission ownership to permission validity under context. In static models, the central question is whether an agent possesses a particular permission. In the proposed architecture, the question becomes whether that permission remains appropriate for the current action, resource, context, and risk state.

This distinction has significant theoretical implications. The supplied literature repeatedly demonstrates that system performance depends on operating conditions and configuration. Solar-energy research addresses changing flux distributions, layout configurations, cost constraints, and optimization alternatives (He et al., 2019; Hayat et al., 2019; Rizvi et al., 2021). Although these studies do not address cybersecurity, their methodological relevance lies in demonstrating that complex systems should not be evaluated using a single fixed configuration. Dynamic authorization follows the same systems principle.

A major practical implication is the need to separate risk estimation from authorization enforcement. An AI model may identify unusual activity, but the security architecture should determine what action follows from

that signal. If an AI model can directly grant permission, model error becomes an authorization vulnerability. If the model instead supplies evidence to a deterministic policy engine, the architecture retains a second line of defense. Bhushan et al. provide a directly relevant example by combining AI-based risk scoring with policy enforcement and differentiated approval mechanisms (Bhushan, Pappu, Jadav, and Jaiswal, 2026).

Another important issue is threshold calibration. A threshold that is too low can create excessive false positives and unnecessary approval requests. A threshold that is too high can permit dangerous operations. Therefore, threshold selection should be treated as an empirical optimization problem. Historical authorization outcomes, incident data, operational requirements, and acceptable risk levels should inform future calibration.

The architecture also introduces governance challenges. Dynamic policies can become difficult to audit when they depend on numerous contextual variables. A security administrator must therefore be able to reconstruct why a request was permitted or denied. Every authorization decision should retain an auditable representation of identity, policy version, contextual variables, risk category, decision, and enforcement action.

The primary limitation is that the proposed architecture is conceptual rather than a full production implementation. Its effectiveness depends on the quality of the risk model, correctness of policies, reliability of identity infrastructure, and completeness of monitoring data. A sophisticated authorization engine cannot compensate for an inaccurate risk model or poorly defined security policies.

A second limitation concerns adversarial adaptation. Autonomous agents and attackers may attempt to manipulate the contextual signals used for risk estimation. Consequently, risk indicators themselves require integrity protection. A third limitation is computational overhead. Continuous evaluation may be costly for high-frequency agent interactions. Efficient caching and hierarchical policy evaluation may therefore be necessary.

The proposed framework should also not be interpreted as eliminating human oversight. Instead, it provides a mechanism for allocating human involvement according to risk. This creates a more scalable model in which routine actions remain automated while consequential actions receive stronger review.

## 6. Conclusion

This research proposed a dynamic risk-based access-control architecture for autonomous agentic AI systems. The framework treats authorization as a continuous, context-sensitive decision rather than as a static permission assignment. It integrates identity validation, delegation constraints, contextual assessment, dynamic risk scoring, deterministic policy evaluation, adaptive enforcement, and continuous monitoring.

The central contribution is the formal separation between risk assessment and authorization authority. Risk provides adaptive information, but deterministic policy constraints remain responsible for the final security decision. This architecture supports proportional security responses ranging from automatic approval to restricted execution, escalation, and denial.

The analysis also demonstrates that methodological principles from complex-system optimization can inform the design of adaptive authorization. The supplied literature emphasizes configuration comparison, optimization under changing conditions, and evaluation of competing system objectives (Danish, 2024; He et al., 2019; Rizvi et al., 2021). These principles can be transferred to authorization without treating the underlying application domains as equivalent.

The proposed model is particularly relevant to agentic environments because autonomous agents can operate across multiple resources and delegation layers. Enforcing monotonic delegation and continuous risk evaluation can constrain privilege propagation and reduce the impact of compromised agents. The architecture also provides a practical basis for differentiated human oversight.

Future research should implement the framework in a real agentic-AI testbed and measure authorization accuracy, latency, privilege containment, false-positive rates, policy consistency, and resilience against adversarial manipulation. Comparative evaluation of static, contextual, and fully adaptive policies would provide stronger empirical evidence. Further research should also investigate automated threshold calibration, formal verification of policy invariants, and scalable authorization for multi-agent environments.

Ultimately, dynamic risk-based access control provides a pathway toward autonomous AI systems that can retain operational flexibility without treating autonomy as unrestricted authority. The most secure architecture is therefore not one that eliminates autonomy, but one that continuously determines the boundaries within which autonomy may safely operate.

## References

1. M. B. Hayat, D. Ali, K. C. Monyake, L. Alagha, and N. Ahmed, “Solar energy-A look into power

generation, challenges, and a solar-powered future," Int. J. Energy Res., vol. 43, no. 3, pp. 1049–1067, Mar. 2019.

2. Y. L. He, K. Wang, Y. Qiu, B. C. Du, Q. Liang, and S. Du, "Review of the solar flux distribution in concentrated solar power: Non-uniform features, challenges, and solutions," Appl. Therm. Eng., vol. 149, pp. 448–474, Feb. 2019.

3. W. M. Hamanah, A. Salem, M. A. Abido, A. M. Qwbaiban, and T. G. Habetler, "Solar power tower drives: A comprehensive survey," IEEE Access, vol. 11, pp. 83964–83982, Mar. 2023.

4. J. Coventry and J. Pye, "Heliostat cost reduction–where to now?," Energy Procedia, vol. 49, pp. 60–70, 2014.

5. M. S. S. Danish, "A framework for modeling and optimization of data-driven energy systems using machine learning," IEEE Trans. Artif. Intell., vol. 5, no. 5, pp. 2434–2443, May 2024, doi: 10.1109/TAI.2023.3322395.

6. A. Pfahl, "Progress in heliostat development," Sol. Energy, vol. 152, pp. 3–37, Aug. 2017.

7. A. Pfahl, "Survey of heliostat concepts for cost reduction," J. Sol. Energy Eng., vol. 136, no. 1, p. 014501, Feb. 2014.

8. A. A. Rizvi, S. N. Danish, A. El-Leathy, H. Al-Ansary, and D. Yang, "A review and classification of layouts and optimization techniques used in design of heliostat fields in solar central receiver systems," Sol. Energy, vol. 218, pp. 296–311, Apr. 2021.

9. K. Varun, U. C. Arunachala, and D. N. Elton, "Trade-off between wire matrix and twisted tape: SOLTRACE® based indoor study of parabolic trough collector," Renew. Energy, vol. 156, pp. 478–492, Aug. 2020.

10. X. Wei, Z. Lu, Z. Wang, W. Yu, H. Zhang, and Z. Yao, "A new method for the design of the heliostat field layout for solar tower power plant," Renew. Energy, vol. 35, no. 9, pp. 1970–1975, Sep. 2010.

11. W. Li, D. Deka, R. Wang, and M. R. A. Paternina, "Physics-constrained adversarial training for neural networks in stochastic power grids," IEEE Trans. Artif. Intell., vol. 5, no. 3, pp. 1121–1131, Mar. 2024.

12. B. Bhushan, K. Pappu, V. Jadav and N. Jaiswal, "Risk-Adaptive Authorization for Agentic AI Systems," SoutheastCon 2026, Huntsville, AL, USA, 2026, pp. 1-6, doi: 10.1109/SoutheastCon63549.2026.11476129.